



How To Check Solar Systems For Efficiency And Long Life

The market for solar systems is growing exponentially due to the rising demand for solar energy worldwide. Unlike other forms of energy sources, solar equipment does not require expensive and consistent maintenance. But for optimised outcomes, it is necessary to ensure that the equipment is working properly at all times. This article discusses the related parameters and several ways of testing the solar equipment

Ayushee Sharma

With the rising demand for solar energy, the market for solar systems is growing exponentially. More and more manufacturers and businesses are considering it as a profitable sector in the long run. New solar systems with better efficiency, performance, and longevity are being launched in the market nowadays. Unlike other forms of energy sources, solar equipment does not require expensive and consistent maintenance. According to a report from Market Research Future, the solar panels market is expected to grow at a CAGR of 20.18 per cent until 2023.

But for optimised outcomes, it is necessary for industries, homeowners, and commercial businesses to ensure that their

equipment is working properly at all times. This makes photovoltaic (PV) testing services mandatory to ensure safety and performance, especially for new products.

It is also important to make sure that common standards for solar panel certification or photovoltaic testing like those established by the International Organization for Standardization (ISO), International Electrotechnical Commission (IEC), American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE), and the like are met. This article discusses the related parameters and different ways of testing solar equipment.

About photovoltaic system

Any photovoltaic system that converts solar energy to power has a few basic components—photovoltaic modules, battery or batteries, charge regulator or controller, and an inverter. A charge controller is necessary for a standalone system for regulating the voltage and current coming from the solar panels and going to the battery so as to prevent the battery from overcharging. An inverter converts direct current (DC) generated in the process to alternating current (AC).

A solar cell or photovoltaic cell is the basic unit of any solar energy generation system. It is a

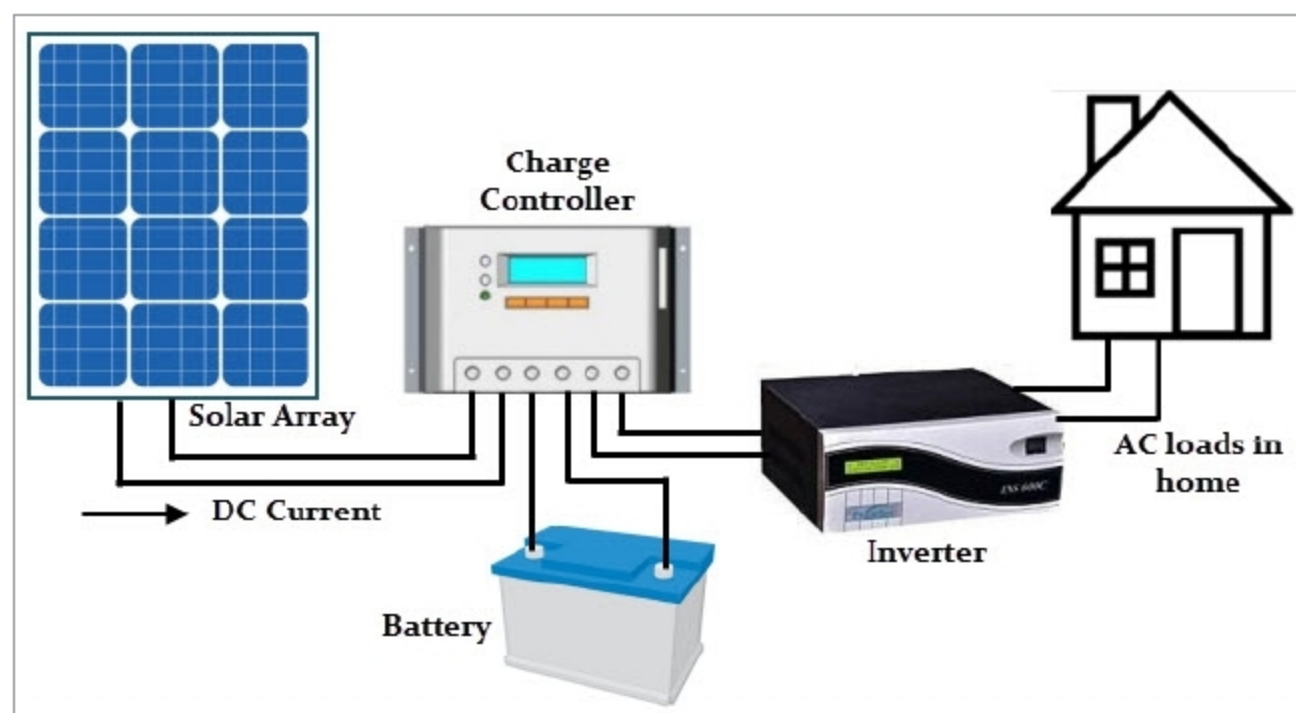


Fig. 1: Basic working of a solar energy power generation system (Credit: www.electronicshub.org)